A Survey on Classification of the Human Brain for Diagnosis of Alzheimer's Disease based on ML, DL Models.

ABSTRACT

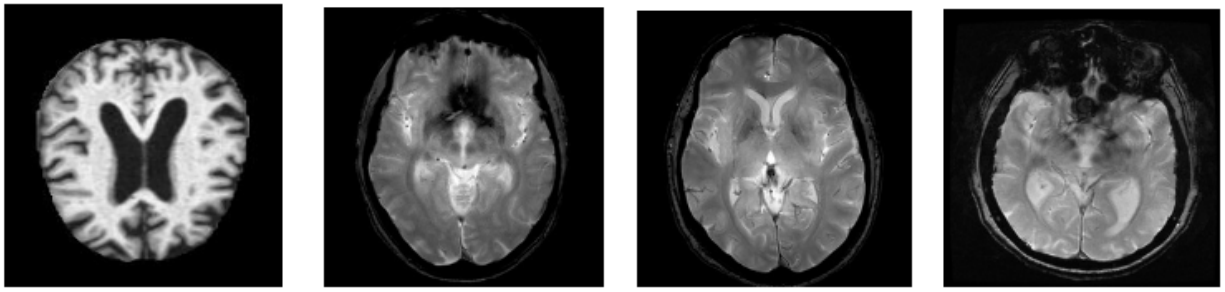
Over the decades, technological advancements, including neuroimaging techniques and biomarker identification, have provided deeper insights into Alzheimer's disease progression. Alzheimer's disease (AD) is an incurable neurodegenerative disease mostly experienced by people who are in the age of early 60s, explicitly impairing detailed mental analysis. The main reason for memory loss in Alzheimer's disease (AD) patients is correlated with progressive damage to the brain, particularly in regions crucial for memory and cognitive function. It has emerged as a significant global health concern with profound significance for individuals, families, and healthcare systems. According to the World Health Organization (WHO), the global prevalence of individuals diagnosed with "dementia" is estimated to be 55 million, with projections indicating an increase to 139 million by the year 2050. The increasing number of people with Alzheimer's disease and the limited availability of effective diagnostic tools make the detection and categorization of Alzheimer's a complex field of research. In recent years, there has been a notable upswing in initiatives directed towards the identification of Alzheimer's disease and addressing its progression. Using computer-based analysis models with machine learning and artificial intelligence has the potential to help us explore different methods of studying the brain. Alzheimer's disease detection is facilitated by utilizing fundamental machine learning architectures such as support vector machines, decision trees, and ensemble models and deep learning models like CNNs. In this paper, we have analyzed a few existing works for AD detection and classification based on their classification performance and conducted a comparative assessment of more than 150 publications related to Machine Learning and Deep Learning across datasets like ADNI, OASIS that have been published since 2015. We have provided detailed summaries of the existing research articles, including thorough observations and also compared the performance and results. Finally, we presented a detailed analysis of the challenges associated with Alzheimer's disease (AD) classification and discussed prospective research directions.

Keywords: Alzheimer's Disease (AD), Neuroimaging Techniques, Biomarker Identification, Cognitive Impairment, Global Health Concern, World Health Organization (WHO), Dementia Prevalence, Support Vector Machines, Decision Trees, Ensemble Models, Deep Learning, Convolutional Neural Networks (CNNs), ADNI Dataset, OASIS Dataset.

1. INTRODUCTION

Alzheimer's disease (AD) is a neurological condition affecting the brain, leading to the degeneration of neuronal cells. While the exact origin of AD remains unclear, individuals with AD experience cognitive decline, including issues with memory and space-time perception. Annually, 10–15% of patients diagnosed with MCI progress to AD dementia [1]. Memory loss is a distinctive symptom of AD, leading to instances where individuals may become disoriented, lose their way, and forget family and friends [2]. In clinical practice, the early identification and categorization of Alzheimer's disease are crucial. Numerous researchers have suggested various methods for classifying Alzheimer's using diverse information sources [3], [4]. Mild cognitive impairment (MCI) functions as a transitional stage between normal cognition (NC) and the initial phases of Alzheimer's disease (AD), signifying an early developmental stage in the progression toward AD [5], [6],[7]. Mild Cognitive Impairment (MCI) is divided into two stages known as Single-Domain MCI (SMCI) and Multiple-Domain MCI (PMCI) [8],[9]. Late Mild Cognitive Impairment (LMCI) and Early Mild Cognitive Impairment (EMCI) are two classifications based on the timing and severity of cognitive decline [10]. Although previous research has provided significant contributions to Alzheimer's disease (AD) diagnosis, there are persistent challenges. Some approaches have constrained the issue to binary classification, restricting their capacity to adequately handle multiple categories [11],[12],[13],[14]. There has been a significant increase in research focused on utilizing neuroimaging for computer-aided classification of Alzheimer's disease (AD) and its early stage, mild cognitive impairment (MCI). This surge in these classification studies has prompted our overview, driven by the need to address the lack of literature summarizing various AD-related features extracted from neuroimaging data and classification algorithms. This work presents three main contributions. We first collected, reviewed and summarized the different studies using Machine Learning as well as Deep Learning algorithms on MRI scanned images for AD classification across datasets like ADNI, OASIS [15] . In particular, we reviewed their validation procedures and image preprocessing procedures, classification architectures and evaluation procedures dedicated to ML & DL. Machine learning provides a systematic methodology for creating advanced, automated, and unbiased classification frameworks to analyze high-dimensional data and has the capability to learn intricate and nuanced patterns of change across diverse imaging modalities [16],[17] whereas Deep Learning presents a systematic approach for constructing sophisticated, automated, and objective classification frameworks when dealing with high-dimensional data, it’s strength lies in the ability to discern complex and subtle patterns of change across diverse imaging modalities [18],[19].We studied the influence

of key components on the classification accuracy, we compared Deep Learning approaches (CNN in particular) to ML approaches majorly based on SVM across datasets like ADNI and OASIS. . In Fig.1, a comparison of brain images is shown between CN, AD, MCI, EMCI for a particular slice position.



(a) CN (b) AD (c) MCI (d) EMCI

Fig1. Comparison of Brain MRI

1.1. AD Classification overview

Over the past decade, classification frameworks have been used successfully to analyze complex patterns in neuroimaging data with a view to the classification of different subjects. A classification framework consists of four major components: data collection, image pre-processing, feature extraction and selection, classification algorithm. Various strategies have been employed by different researchers in the classification of Alzheimer's disease (AD), with a commonly adopted approach outlined in the block diagram depicted in Fig 2.

Data Collection

The primary and crucial stage in classifying Alzheimer's disease (AD) is to secure ample and accurate data. The process of data acquisition includes collecting various elements like brain images, the patient's medical history, genetic information, and more. Researchers commonly obtain data from a range of clinics and hospitals, often relying on widely used online datasets such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS).By incorporating data from these sources, researchers ensure a robust foundation for subsequent analyses and classifications, contributing to the advancement of understanding and addressing the complexities associated with Alzheimer's disease.

Image pre-processing

To elevate the quality of Magnetic Resonance Imaging (MRI) data and mitigate the influence of noise, a meticulous image pre-processing method was implemented [20].

This intricate process not only refines the raw data but also plays a crucial role in preparing it for subsequent analytical steps. Standardizing the data is deemed indispensable to ensure uniformity across all data points, facilitating a consistent scale throughout the dataset [21]. Within the domain of brain images, fundamental pre-processing techniques constitute essential steps, encompassing pivotal procedures such as skull stripping, de-noising, and segmentation. Among these, skull stripping emerges as a prominently adopted approach, directed at the removal of extraneous pixels that contribute minimally to the accurate representation of the brain in the images. This particular step holds significant importance in refining the data by eliminating unwanted elements and creating a strong foundation for subsequent analyses. The elimination of unnecessary details not only enhances the overall quality of the brain images but also streamlines the subsequent analytical processes, contributing to a more precise and insightful understanding of the complex structures within the images.

Feature Extraction and Selection

Brain images serve as repositories of vast information, yet discerning the relevant fraction crucial for classification demands a specialized approach. The intricate process of extracting pertinent information from these images, known as feature extraction, employs sophisticated methods including partial least squares, principal component analysis, and Gaussian mixture model [22]. These techniques systematically unravel the intricate patterns and nuances embedded within the images.

Post-extraction, the imperative arises for feature selection, a crucial step to preserve only the most pivotal elements. Commonly employed techniques for feature selection encompass support vector machine-recursive feature elimination, t-tests, and statistical dependency analysis [23] In essence, this two-step approach orchestrates a meticulous curation of features, ensuring that subsequent classification endeavors concentrate on the most meaningful elements. This optimization not only enhances the efficiency of the analysis but also elevates the accuracy of the overall classification process, facilitating a more nuanced and precise understanding of the patterns inherent in brain images.

Classification

Classification represents the culmination of the analytical process, providing a decisive outcome from various potential scenarios through the deployment of a classifier. In the specific domain of Alzheimer's Disease (AD) classification using brain images, an array of well-established classifiers comes into play. These classifiers include advanced techniques such as support vector machines, random forests, artificial neural networks (ANN), K-nearest neighbors, and sophisticated neural network architectures like convolutional neural networks (CNN) and recurrent neural networks (RNN). Each classifier brings its unique advantages and approaches to the task, playing a crucial role in the intricate process of accurately identifying and differentiating patterns within brain images for AD classification. These classifiers contribute significantly to

advancing the understanding and diagnosis of Alzheimer's disease, offering diverse tools to uncover intricate patterns and nuances within brain images.

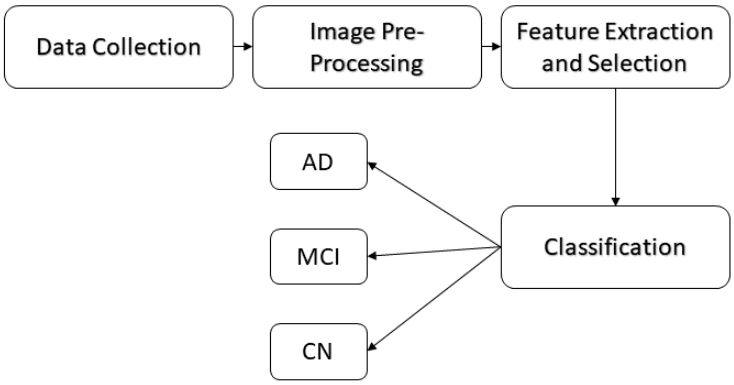


Fig 2. AD Classification Approach.

1.2. Selection Criteria

In the process of organizing this survey paper spanning from 2015 to 2023, we selected studies that encompassed the most significant and beneficial research pertaining to the classification of Alzheimer's disease (AD).We searched prominent papers in the field from IEEE, ResearchGate, GoogleScholar and Semantic Scholar and excluded the studies which did not use accuracy measures for classification performance. This resulted in a total of approx. 170 papers. Our focus was mainly on studies that directly contributed to the classification of AD, exploring innovative methods for data collection, preparation, feature extraction, and subsequent utilization for classification. We included papers based on both Machine Learning and Deep Learning models that mainly focused on SVM and CNN. We ensured the inclusion of papers from esteemed journals and conferences, prioritizing those that had a substantial impact and employed diverse methodologies. Our objective was to incorporate studies that genuinely contributed novel insights to our comprehension of AD classification. Through this approach, we aspire to provide readers with a comprehensive overview of the varied and influential research that has molded this field over the past eight years.

2. MRI DATASETS AND ANALYSIS

The most important step for AD classification is to acquire adequate and accurate data. Acquisition of data for AD classification indicates to obtain brain images, patient's medical history, genetic information. The assessment framework for three-dimensional (3D) cross-sectional brain MRI data is employed for the categorization of individuals with Alzheimer's disease (AD) and the segmentation (a process that involves partitioning the brain into different regions) of brain tissue types, including cerebrospinal fluid (CSF), gray matter (GM), and white matter (WM) [24]. This segmentation is crucial for studying normal brain development, aging, and identifying abnormalities associated with neurological disorders. They are instrumental in detecting and monitoring neurological disorders such as Alzheimer's disease, multiple sclerosis, and brain tumors. Publicly available datasets such as Open Access Series of Imaging Studies (OASIS), Alzheimer's Disease Neuroimaging Initiative (ADNI), Kaggle, EEG, AIBL are popularly used for segmentation of brain MRI and AD diagnosis and acquired through Magnetic Resonance Imaging, a non-invasive imaging technique that provides high-resolution, three-dimensional images of the brain.

2.1 Public Datasets for Brain MRI

2.1.1. ADNI

The ADNI dataset is a comprehensive resource that includes various types of imaging data, such as structural and functional MRI, positron emission tomography (PET), and other biomarker measurements[25]. It is specifically designed for studying Alzheimer's disease and related disorders. ADNI's fundamental objective is the development and validation of biomarkers for Alzheimer's, spanning various data types, including structural and functional magnetic resonance imaging (MRI and fMRI), positron emission tomography (PET), genetic data, cerebrospinal fluid (CSF) biomarkers, and clinical assessments. Over the years, ADNI has progressed through multiple phases, including ADNI-1, ADNI-GO (Grand Opportunities), ADNI-2, and ADNI-3, each contributing to different facets of Alzheimer's research. Longitudinal follow-up of participants allows for the comprehensive study of changes in brain structure, function, and cognitive decline over time. Also, we have to note that portions of the ADNI dataset are publicly accessible, enabling researchers to delve into the wealth of information it offers. However, access to certain sensitive data, such as genetics or

detailed clinical information, may require specific permissions and adherence to ethical guidelines.

2.1.2. OASIS

The Open Access Series of Imaging Studies (OASIS) dataset is a significant resource in the realm of neuroimaging and is established with the goal of advancing research on aging, Alzheimer's disease, and related neurological disorders, OASIS has become a valuable tool for the scientific community[26].

OASIS includes a diverse range of cross-sectional MRI data, featuring brain images from both healthy individuals and those diagnosed with Alzheimer's disease. The dataset contains multiple modalities, allowing for a detailed examination of brain structures and functions. This diversity helps us to explore various aspects of neuroanatomy and pathology, supporting investigations into the impacts of aging and neurodegenerative conditions. One characteristic of the OASIS dataset is its focus on open access, promoting transparency and collaboration within the research community. We commonly leverage the OASIS dataset for tasks such as brain segmentation, which involves partitioning the brain into distinct regions such as gray matter, white matter, and cerebrospinal fluid [24]. This segmentation process is the key for studying structural changes in the brain associated with aging and neurodegenerative diseases. Additionally, OASIS contributes to the evaluation of novel imaging techniques, providing a baseline for the development and validation of advanced neuroimaging methods. The extended nature of some OASIS data allows for the investigation of changes in brain structure over time. This temporal dimension is crucial for understanding the progression of neurological conditions, contributing insights into the dynamic aspects of brain health and disease.

In summary, the OASIS dataset serves as a valuable repository for brain MRI data, fostering research on aging, Alzheimer's disease, and related neurological disorders. Its open-access nature and diverse content make it an important resource for advancing our understanding of brain structure, function, and pathology. We can access the OASIS dataset to conduct a wide array of studies, ultimately contributing to progress in neuroscience and clinical practice.

3. A REVIEW ON ML & DL MODELS

3.1 Machine Learning

3.1.1 Supervised Learning

KNN

The K-Nearest Neighbour (KNN) algorithm, a versatile data mining technique widely applied in image processing [27], serves dual purposes in both classification and regression problems,including the classification of Alzheimer's disease. In the K-NN classification approach, an object is assigned to a specific class by considering the majority classes among its K neighbors, with the value of K being a positive integer that determines the number of neighbors for polling [28]. This value of K is a critical parameter for the algorithm, defining the number of nearest neighbors considered for classifying the data. We train the algorithm on a labeled dataset to distinguish between individuals with Alzheimer's disease and those without. Choosing an appropriate value for the 'k' parameter in KNN can be done through cross-validation on the training set to find the value that gives the best performance. While KNN can be used for classification tasks, its performance may vary depending on the nature of your data and the characteristics of Alzheimer's disease.

The implementation of the KNN algorithm involves several steps [29]:

- Determining the value of K, representing the number of nearest neighbors to be considered. Higher K values lead to increased complexity, requiring the resolution of conclusions through the Euclidean equation. Choosing a value of K=1 is often preferred for its simplicity and quicker classification determination.
- Calculating the squared Euclidean distance individually for each data point.

$$d = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} \tag{1}$$

Where x_i and y_i are two feature vectors used in classification

- Sorting the results in ascending order.
- Aggregating the classification results based on the varying values of K.

Algorithm

Input: Extracted feature set (A), training object (x), testing object (y), category label set (C)

Output: Category c_x of test object x, c_x belongs to class C

```
begin
   for each y belongs to A
         calculate the Distance A(y, x) i. e . between y and x
    end for
         select the subset N from the data set A,
          N contains k training samples which are the k
          nearest neighbours of the test samples x
    calculate the category of x:
    c_x = arg max Σ_{y∈N} I(c = class(y))
  end
end
```

Naive Bayesian

The Naive Bayes algorithm, a classification technique in both data mining and machine learning, relies on the Bayesian theorem.[28] This method is employed to determine the probability of an attribute by considering other known probabilities associated with the attribute. Gaussian probability densities are utilized for modeling the data within the framework of Naive Bayes. The algorithm employs Gaussian probability densities to model data. This entails assuming that the distribution of the data conforms to a Gaussian (normal) distribution, characterized by a bell-shaped curve. The use of Gaussian probability densities allows Naive Bayes to handle continuous data effectively, providing a statistical framework for expressing the likelihood of attribute values. This integration of probabilistic modeling enhances the algorithm's ability to make well-informed predictions and classifications in the context of Alzheimer's disease. Consequently, Naive Bayes emerges as a versatile and widely adopted tool for Alzheimer's classification, demonstrating its efficacy, efficiency, and simplicity in managing diverse datasets within the healthcare domain[30].

Likelihood

Class Prior Probability

$$P(c \mid x) = \frac{P(x \mid c)P(c)}{P(x)}$$

Posterior Probability

Predictor Prior Probability

$$P(c \mid \mathrm{X}) = P(x_1 \mid c) \times P(x_2 \mid c) \times \cdots \times P(x_n \mid c) \times P(c) \tag{2}$$

Random Forest

The Random Forest algorithm is a powerful classification technique widely used in data mining and machine learning. Unlike Naive Bayes, which depends on probabilistic modeling, Random Forest works as an ensemble learning method [31]. It builds multiple decision trees during training, with each tree independently constructed using a random subset of features and training data. This ensemble approach allows Random Forest to reduce overfitting and enhance predictive accuracy. Being versatile, it is capable of handling both categorical and numerical attributes by randomly selecting subsets of features for each tree. Because Alzheimer's datasets can be quite diverse, containing a mix of different kinds of information like categories and numbers, Random Forest stands out [32]. It's good at handling this mix of information smoothly and can easily include various features such as genetic details, brain imaging data, and medical evaluations, making it a well-rounded approach to categorizing or classifying information. Trained on labeled datasets containing information about patients with and without Alzheimer's disease, the model can predict the likelihood of Alzheimer's based on new input data. This predictive capability is pivotal for early diagnosis and facilitates proactive treatment planning.

$$RFfi_i = \frac{\sum_j normfi_{ij}}{\sum_{j \in all\ features, k \in all\ trees} normfi_{jk}} \tag{3}$$

- RFfi sub(i)= the importance of feature i calculated from all trees in the Random Forest model
- normfi sub(ij)= the normalized feature importance for i in tree j
- T = total number of trees

Algorithm

Input: Parameters set to produce RF classification of a training set [A = (x(1), y(1)), . . . ,(x(n), y(n))], x(i) and y(i): training data, F: different features set f and number of trees in forest b.
Output: The prediction metrics

```
function Random Forest (A, F)
  R ← ∅
  for i ∈ 1, . . ., b do
     A(i) ← A bootstrap sample from A
     h_i ← Random Tree Learn (A(i), f)
     R ← R ∪ {r(i)}
   end for
   return R
end function
function Random Tree Learn, A (H, f)
At each node:
    f ← very small subset of f estimated from 100 of (x,y)
       Split on best feature in f
    Return The learned tree
end function
```

Decision Tree

Decision tree works by recursively splitting the dataset into subsets based on the most informative features, creating a hierarchical decision structure. It is a supervised machine learning model that mimics the structure of a tree, where each internal node represents a decision based on a feature, each branch represents the outcome of that decision, and each leaf node represents the final prediction or classification[33].In the context of classification, decision trees are particularly effective. They are used to classify instances into different categories based on a set of input features. The decision-making process involves evaluating the values of features at each internal node and traversing the tree until a leaf node is reached, providing the final prediction. The construction of a decision tree involves selecting the most informative features for splitting the data at each node [34]. The goal is to create decision rules that effectively partition the data into homogeneous subsets, leading to accurate and interpretable predictions. In Alzheimer's disease classification, decision trees can be utilized to analyze patient data and identify key factors influencing disease presence or progression by understanding the decision rules generated by the model.

$$E(S) = -p_{(+)}\log p_{(+)} - p_{(-)}\log p_{(-)} \tag{4}$$

- p+ is the probability of positive class
- p– is the probability of negative class
- S is the subset of the training example

Algorithm

Input: Extracted feature sets (C)
Output: The prediction metrics

Creating root node (based on feature set)
All in same class C
Root node = ***single − node tree with label*** = ***C***
if Attribute is empty
then Root node = single node tree with label = most common value of target_attribute in Examples
else
A = element in attributes that maximizes information gain(Examples A)
A = Decision attribute root node
For each possible value v of A
Adding Branch, testing for A
V = ***subset with A = v***
If *Series_v is empty*
Then below branch add leaf with label = most common value of target
else
Below branch add subtree
return
Root node

Support Vector Machine (SVM)

SVM is a supervised machine learning algorithm that analyzes the data used for classification and regression. SVM performs well in handling high feature space and also has flexibility in choosing similarity functions. functions. In SVM classification, the kernel function is very important since it corresponds to an inner product in some expanded feature space. The kernel function plays the role of dot product in feature space [35]. SVM has been used in many AD classification studies and obtained competitive balanced accuracies. In order to classify data that is not linearly separable, kernel functions are employed to transform the data into higher dimensions and several SVM variants have been suggested to enhance its performance in terms of generalization capability and training time. Different kernels have been used with SVM for classification of Alzheimers. Various researchers have used linear kernel with SVM to classify Alzheimer data. It is crucial for effectively classifying linearly separable data, providing simplicity, interpretability, and computational efficiency. This is due to the fact that in a linear kernel, there is no kernel parameter to tune and it is simple to use [36],[37],[38] . Support Vector Machines with Linear Kernel (svmLinear) constructs hyperplanes high dimensional data. It is based on the concept of decision planes that define decision boundaries and perform linear regression in the high-dimension feature space.

Linear SVM

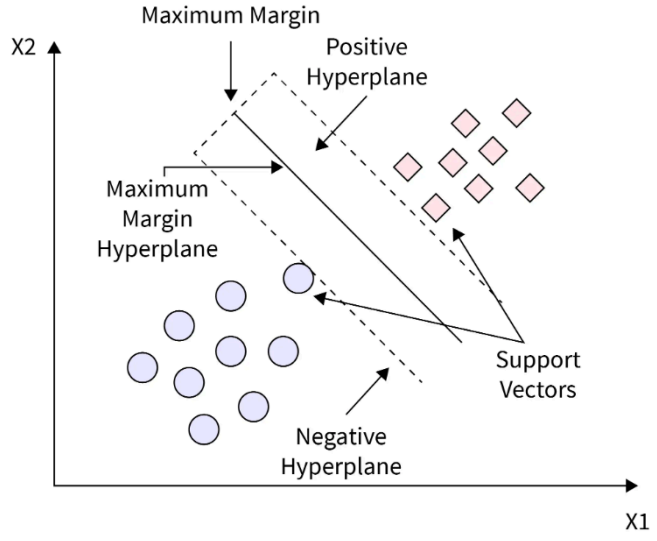


Fig 3. Linear SVM

In the context of Alzheimer's disease classification, as depicted in Fig. 3, the bold line emerges as the optimal separator among the three lines, delineating distinct classes such as AD (Alzheimer's Disease), CN (Cognitively Normal), MCI (Mild Cognitive

Impairment), etc., through the maximum marginal hyperplane. The linear equation characterizing this hyperplane takes the form Y = AX + b, where X represents the feature vector, A is the coefficient vector, and b is a constant. This predictive hyperplane becomes pivotal in classifying individuals into specific diagnostic categories related to Alzheimer's disease. The points located along the dotted lines, identified as support vectors, play a critical role in the classification process. SVM demonstrates its capacity to model nonlinear decision boundaries within the original feature space, a crucial aspect in the classification of individuals into different Alzheimer's disease-related classes, achieved through the application of the kernel trick.

Multi-Class SVM

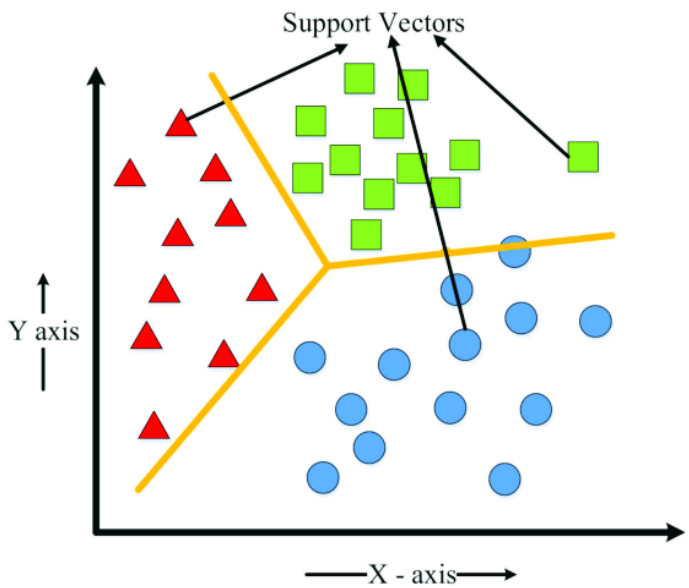


Fig 4. Multi Class SVM

Some researchers have leveraged the effectiveness of multi-class SVM. This approach serves to assess and compare the predictive value of both well-established and non-invasive, easily accessible candidate variables. The aim is to classify participants into distinct cognitive states, including healthy cognition, subjective cognitive decline, amnestic mild cognitive impairment, and mild Alzheimer's disease [39].

In the context of multi-class classification, as illustrated in Figure 4, diverse strategies such as pairwise and one-versus-all are employed to transform the multi-class problem into binary-class classification . Moreover, the literature reveals additional formulations of binary classification specifically tailored for streamlined handling of compact multi-class classification problems.

Non-Linear SVM

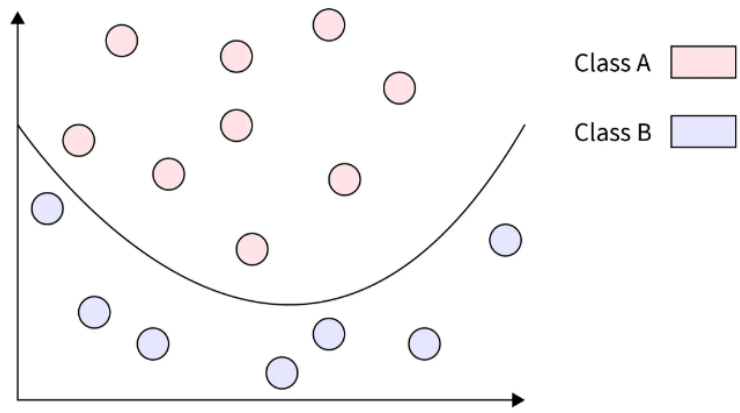


Fig 5. Non-Linear SVM

While some researchers used Non-Linear SVM that employs kernel functions for complex data, enhancing classification accuracy by mapping input to higher-dimensional spaces, improving pattern recognition. Linear SVM is suitable for linearly separable data, where a hyperplane can effectively divide classes, while non-linear SVM is designed to handle complex, non-linear relationships within the data using kernel transformations. Non-linear SVM uses kernel functions to accommodate complex, non-linear relationships between features, enabling better classification in higher-dimensional spaces.

Triple SVM

In the context of Alzheimer's disease classification, the Triple SVM classifier is employed as a sophisticated ensemble method. This approach utilizes three Support Vector Machines (SVMs) to enhance classification accuracy by combining their individual strengths and capitalizing on ensemble synergy. The triple SVM classifier proves advantageous in preventing information loss, especially when considering class data against each other, and it exhibits enhanced performance on datasets like IQ-OTH or ADNI. Unlike traditional SVM classifiers, the triple SVM stands out for its ensemble approach, employing three individual SVMs that collectively contribute to the classification process. In contrast to linear SVMs, suitable for linearly separable data, the triple SVM extends its applicability to handle intricate patterns and complex relationships within the data through the ensemble's collective decision-making, surpassing the limitations of a single decision boundary [40].

The linear classifier is depends on the linear discriminant function $f(F_{ea})$ which is expressed in equation

$$f(F_{ea}) = w^T F_{ea} + b$$

Where, w is the weight vector and b is the bias value

There are still many kinds of varieties in SVM classifiers including RBF SVM, LibSVM and HPT-TSVM. Radial Basis Function Support Vector Machine is a type of Support Vector Machine (SVM) that utilizes a Radial Basis Function (RBF) or Gaussian kernel. The RBF kernel is commonly used in SVMs for handling non-linear relationships in the data. The RBF kernel works by transforming the input data into a higher-dimensional space, where nonlinear relationships can be better captured. It measures the similarity (or distance) between data points in this higher-dimensional space. The RBF kernel is particularly effective in scenarios where the decision boundary between classes is not linear and exhibits complex patterns. The RBF kernel allows the algorithm to create a decision boundary that can be highly flexible and adapt to intricate relationships within the data. Whereas, LibSVM is widely used in pattern recognition applications due to its efficiency and versatility. It supports multi-class classification, cross-validation, and grid search for parameter tuning. LibSVM is implemented in C and has interfaces for various programming languages, making it accessible and widely adopted in both academic research and practical applications [41]. Hyperparameter tuning is a critical aspect of optimizing the performance of machine learning models, and it is equally important for the Twin Support Vector Machine (TSVM), including its hyperparameter tuning variant

HPT-TSVM

HPT-TSVM involves adjusting the parameters that are not learned from the data but are set prior to the training process to enhance the model's predictive accuracy. These hyperparameters may include the regularization parameter, kernel parameters, and other configuration settings [42].

Mathematical Representation Of HPT-TSVM

Provided training vectors $X_n \in R^q$, where n = 1, ..., p in two classes, and a vector $V \in \{1, -1\}^i$. The primary objective is to discover the optimal weight, $W \in R^q$, and bias, $b \in R$, in a manner such that the prediction is dictated by the sign ($w^T + b$). The resolution of the Support Vector Classifier (SVC) resides in the initial problem outlined below.

$$\min_{w,b,\alpha} \frac{1}{2} w^T w + D \sum_{q=1}^{t} \alpha_q \tag{5}$$

Increasing the penalty for misclassifications aims to maximize the margin, operating under the assumption that $||w||^2 = w^T w$. Achieving perfect classification necessitates that the value $V_q(w^T \rho(X_n)+b)$ equals or exceeds 1. The strength of the penalty, denoted by D, plays a crucial role. The corresponding dual problem of the primal solution is presented below.

$$\min_{\beta} \frac{1}{2} \beta^T S \beta - \epsilon^T \beta \tag{6}$$

subject to $V^T \beta = 0$ where $0 \leq \beta_i \leq 0$, i=1,……. P

Here, e signifies the vector, and V is an m × m positive definite matrix. The term β_i represents the dual coefficients. This representation involves mapping the training vectors into a higher-dimensional space, denoted as D-space. Consequently, the output of the objective function for sample X is illustrated in the below equation.

$$\sum_{i \in SV} V_i \, \beta_i K(X_n, X) + b \tag{7}$$

As discussed, different variants of SVM have been employed for classification of Alzheimers. They are well-suited for Alzheimer's Disease (AD) classification due to their proficiency in handling high-dimensional data, a common characteristic of AD datasets such as neuroimaging data. The ability of SVMs to capture non-linear relationships is particularly beneficial in AD classification tasks, where complex patterns are often present. By employing kernel functions, such as the radial basis function kernel, SVMs can effectively model intricate relationships within the data. Additionally, SVMs exhibit robustness to overfitting, a crucial feature when working with medical data where overfitting can lead to misleading results. The versatility, non-linear modeling capabilities, and robustness make SVMs a compelling choice for AD classification, contributing to their widespread use in research and clinical applications related to Alzheimer's Disease.Moreover, other than the existing models, some novel variants of SVM also need to be developed for Alzheimer’s disease.

3.1.2. Unsupervised Learning

K Means

K-means clustering is a popular unsupervised machine learning algorithm used for partitioning a dataset into distinct groups or clusters based on similarity patterns within the data. The algorithm works by iteratively assigning data points to clusters and updating the cluster centroids until convergence. The number of clusters, represented by the 'k' parameter, is predefined by the user. K-means minimizes the within-cluster sum of squares, aiming to create clusters where the variance within each cluster is minimized [43]. Each cluster is characterized by a centroid, which is the mean of the data points belonging to that cluster. K-means is versatile and applicable to various domains, such as customer segmentation, image compression, and pattern recognition. While sensitive to initial cluster centers, K-means is computationally efficient and particularly useful when dealing with large datasets. It provides a straightforward approach to grouping similar data points, making it a valuable tool in exploratory data analysis and clustering applications.

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \underbrace{\left\| x_i^{(j)} - \mu_j \right\|^2}_{\text{Euclidian distance}}$$

No. of clusters → k; No. of data points → n; data point belonging to cluster j → $x_i^{(j)}$; centroid → μ_j; cluster → j; data point → i

(8)

Gausian mixture model

Gaussian Mixture Model (GMM) is a probabilistic model widely employed in statistical modeling and machine learning. It assumes that the observed data is a mixture of several Gaussian distributions, each with its own set of parameters including mean, covariance matrix, and weight. They are particularly useful in scenarios where the data is not easily categorized into distinct clusters and may exhibit complex patterns and find applications in various domains, such as clustering, density estimation, and anomaly detection. In [44], researchers implemented Gaussian mixture model (GMM) clustering for its ease of implementation, generalizability to common cluster shapes, and probabilistic cluster membership interpretation. In Alzheimer's disease classification, Gaussian Mixture Models (GMMs) are employed to model the distribution of cognitive features extracted from brain imaging or neuropsychological assessments. By representing the heterogeneity in cognitive profiles across individuals with and without Alzheimer's, it can contribute to more accurate classification models, which will help in the early diagnosis and understanding of the disease progression. The probabilistic nature allows for a flexible representation of the complex variations in cognitive data associated with Alzheimer's disease, enhancing the sensitivity and specificity of classification algorithms.

$$N(\mu, \Sigma) = \frac{1}{(2\pi)^{\frac{d}{2}} \sqrt{|\Sigma|}} exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

(9)

- μ = Mean
- Σ = Covariance Matrix of the Gaussian
- d = Number of Features in our Dataset
- x = Number of Data Points

3.1.3 Semi supervised learning

GAN

Alzheimer's Disease classification based on SSGAN involves two models: Generator (Gm) and Discriminator (Dm). These models engage in a competitive dynamic, resembling two players in a minimax game, utilizing the derived function P(Dm, Gm) from the equation[45].

$$\underset{Gm}{\text{Min}}\ \underset{Dm}{\text{max}} \sum_{i+kdata}[\log Dm(i))] + \log 1 - Dm(Gm(a))] \tag{10}$$

The input image is represented by 'i,' and the noise vector is denoted as 'a.' The Generator creates a mapping function for the data space, denoted as Gm(a), based on the prior noise distribution ka a. Both Gm and Dm are fine-tuned to reduce training time, with adjustments made to Dm's constraints and log(1 - D(G(z))) modified to minimize log(D(x)). The objective function of the GAN is as follows:

$$\underset{Gm}{\text{Minmax}}\ \underset{Dm}{} \sum_{i+kdata}[\log Dm(i))] + \log 1 - Dm(Gm(a/j))] \tag{11}$$

The symbol 'j' denotes the labeled image incorporating auxiliary data. The utilization of traditional GAN combined with the loss, as indicated by prior studies, presents advantages. Moreover, the inclusion of the ls1 loss contributes to enhanced performance in the segmentation model. Consequently, the loss technique for the generator, denoted as LsGm, is defined and may be expressed as.

$$Lx_i Gm = \sum_{i+kdata}[\log Dm(i))] + \log 1 - Dm(Gm(a/j))] \tag{12}$$

To prevent overflow, 'i' signifies the input image, and 'e' stands for the observed number. The labeled images of the ground truth Hippocampus are denoted as 'lm.' To offset the losses, values of 1 and 2 were employed with 0.999 and 0.001, respectively. The loss mechanism for the discriminator, denoted as LsDm, is outlined as follows:

$$Lx_i Gm = \sum_{i+j,kdata}[\log Dm(i))] + \log 1 - Dm\left(Gm\left(\frac{a}{j}i\right) + e\right] \tag{13}$$

The optimization process involves refining both the generator and discriminator for Alzheimer's disease classification using a semi-supervised GAN. The optimization is executed through the utilization of the Adaptive Moment Estimation (Adam) algorithm. The input for this process is an MRI image dataset, while the corresponding labeled MRI image serves as the ground truth[46].

3.1.4 Ensemble Learning

Gradient Boost

The application of gradient boosting trees, particularly through the XGBoost library, has demonstrated remarkable success in Alzheimer's disease classification[29]. Leveraging its unsupervised learning approach, XGBoost excels in discerning intricate patterns within diverse datasets. Specifically adapted for Alzheimer's disease classification, XGBoost stands out for its speed and accuracy. Its application extends to efficiently categorizing different stages of Alzheimer's severity, showcasing its stability even with slight adjustments in predictions[47]. This method proves pivotal in enhancing the precision and reliability of Alzheimer's disease classification models[48].

AdaBoost

AdaBoost, as an ensemble learning strategy, has proven effective in Alzheimer's disease classification[49]. Utilizing decision trees with a single level, or Decision Stumps, AdaBoost excels in aggregating weak learners to create a robust classification model. In the context of Alzheimer's disease, AdaBoost's simplicity and effectiveness contribute significantly to improving predictive performance. Its adaptability makes it a valuable tool for refining Alzheimer's disease classification models, providing reliable and accurate results.

Multiple Kernel Learning (MKL)

Multiple Kernel Learning (MKL), particularly through the MKBoost algorithm, showcases promise in Alzheimer's disease classification tasks. Creating a diverse set of base kernels for each dataset, MKL, when applied to Alzheimer's disease classification, enhances accuracy and robustness[50]. The adaptability of MKL techniques proves beneficial in discerning different stages of Alzheimer's severity, contributing to improved classification accuracy. Additionally, the integration of SVM using the LIBSVM toolbox further highlights MKL's versatility and effectiveness in addressing complex Alzheimer's disease classification challenges. The ensemble learning approach, particularly with MKL, emerges as a valuable asset in advancing the accuracy and efficiency of Alzheimer's disease classification models.

Hierarchical Ensemble Learning (HEL):

Hierarchical Ensemble Learning (HEL), a notable characteristic is the deliberate absence of restrictions on the choice of classifiers, providing the flexibility to utilize any available classifier. This encompasses not only various classifiers but extends to implementing both ensemble components using the same method. To illustrate, consider the application of a linear model, specifically Logistic Regression denoted as $\mathbf{f(x)=w \cdot x}$ The training process involves minimizing the following loss function:

$$\min_{\mathbf{w}} \sum_{i}^{T} \log(1 + exp(-r_i(\mathbf{w} \cdot \mathbf{x}_i)) + \|\mathbf{w}\|_2^2 , \tag{14}$$

Here, T represents the number of training images collected from I. The introduction of an auxiliary label r_i is significant, where r_i=1 if the i-th training slice belongs to the c-th class (assuming the training of the c-th binary classifier), and r_i= -1 otherwise. The left term in Equation diminishes when the sign of $w \cdot x$ aligns with r_i while the right term acts as a regularizer to prevent overfitting. Solving this optimization problem is efficiently achieved using the LIBLINEAR package.The two ensemble learning steps can seamlessly replace *x* in Equation with *y* and *z* respectively, facilitating a straightforward training process within the HEL framework.

HEL proves instrumental in enhancing the precision and reliability of classification models. Experimental results show that the HEL framework can obtain notable performance gains with respect to various features and classifiers [51]. The flexibility to utilize various classifiers and seamlessly integrate ensemble components offers a robust approach to discerning different stages of Alzheimer's severity, thereby contributing to improved accuracy in disease classification models.

3.2 Deep Learning

3.2.1 Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) stand out as a powerful tool in the domain of deep learning, particularly when applied to medical imaging and the detailed task of Alzheimer's disease classification[52]. As a subtype of neural networks, CNNs are tailored to process and analyze visual data, making them invaluable in deciphering complex patterns inherent in medical images associated with Alzheimer's disease diagnosis. CNNs showcase their effectiveness primarily in image recognition and classification, precisely the capabilities required for discerning intricate patterns indicative of Alzheimer's disease stages[53]. This technology has proven instrumental in identifying subtle variations in brain structures and abnormalities, contributing to early and accurate diagnosis. The architecture of CNNs, characterized by multiple layers including convolutional layers, allows them to learn detailed features from medical images[54]. In the case of Alzheimer's disease, this includes the identification of specific patterns and anomalies in brain scans that may signify different stages of the disease. Subsequent layers, such as fully connected layers, aid in the classification of these images based on the extracted features[55]. The use of CNNs in Alzheimer's disease classification holds immense promise, not only for accurate diagnosis but also for gaining deeper insights into the progression of the disease. Whether categorizing brain scans into non-demented, very mild demented, mild demented, or moderate demented

classes, CNNs contribute significantly to the understanding and early intervention of Alzheimer's disease.Leveraging various CNN architectures significantly boosts their effectiveness in the classification of Alzheimer's disease. Architectural variants such as VGG, ResNet, Inception, and others introduce unique innovations that empower researchers to customize these models for specific intricacies encountered in Alzheimer's disease research. This assortment of CNN architectures not only amplifies the precision of disease classification but also permits a more nuanced adaptation to the specific challenges inherent in Alzheimer's disease analysis[56][57][58].

The diverse array of CNN architectures plays a pivotal role in refining our understanding of Alzheimer's disease by offering distinct methodologies for capturing and interpreting intricate features within medical images[59]. Different types of CNN designs help researchers better understand Alzheimer's disease by allowing them to identify subtle patterns and anomalies that indicate different stages of the illness. This diverse approach, made possible by using various CNN architectures, improves the sophistication and accuracy of Alzheimer's disease classification[60]. This progress marks a significant step forward in the intersection of deep learning and medical research.

Working of CNN

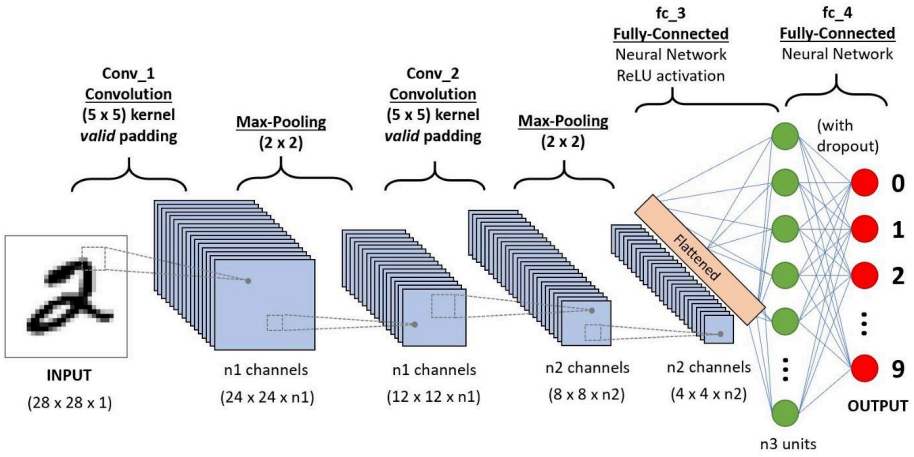


Fig 6. Working of CNN

The convolution layer

The convolution layer, a fundamental component of Convolutional Neural Networks (CNNs), is integral to the classification of Alzheimer's disease. Through the application of filters and convolution operations, CNNs extract distinctive features from medical images, revealing critical patterns indicative of different cognitive states. This process

aids in the identification and classification of Alzheimer's disease stages, contributing to the overall effectiveness of the CNN model[61].

When dealing with a three-dimensional image characterized by dimensions H × W × C (where H is height, W is width, and C is the total channel count), a 3D filter of size F × H F × W F × C is applied. Here, FC represents the number of filter channels, FW denotes the filter width, and FH indicates the filter height. Consequently, the resulting activation map size should be AH × AW, with AH representing activation height and AW representing activation width. The determination of activation height and width values involves the application of specific equations[62].

$$A_H = 1 + \frac{H - FH + 2P}{S}$$
$$A_w = 1 + \frac{W - Fw + 2P}{S} \tag{15}$$

Padding is denoted by P, Stride is represented by S, and when there are n filters, the activation map dimensions should result in $A_H \times A_W \times n$.

The pooling layer:

In the context of Alzheimer's disease classification, the pooling layer plays a crucial role in preserving relevant image characteristics. The pooling layer aims to reduce the dimensions of the feature maps, resulting in fewer parameters to learn and less computational load for the network. Various pooling techniques, including max pooling, average pooling, and global pooling, serve this purpose. Additionally, the activation function introduces non-linearity to the network by applying a non-linear transformation to the given inputs[63]. The most commonly used activation function in the output layer of multi-classifier is SoftMax.. SoftMax calculates relative probabilities, and its general equation is expressed below in Equation.

$$\text{Softmax}\,(z_i) = \frac{\exp(zi)}{\varepsilon jexp(Zj)} \tag{16}$$

In this context, Z represents the values of the output layer neurons, and the exponent serves as a non-linear function. Subsequently, these values undergo normalization and transformation.

The correction layer :

The rectification layer (ReLU layer) enhances the efficiency of Alzheimer's disease classification by introducing non-linearity through an activation function. This layer ensures that neurons produce positive values, allowing the CNN to effectively capture complex relationships within the medical images associated with Alzheimer's disease progression[64]. The commonly used activation function in this case is the ReLU activation function, which is a widely recognized function in CNNs. Different variants of ReLU activation functions, including parametric ReLU, leaky ReLU, exponential linear (ELU, SELU), and concatenated ReLU (CReLU), exist. Leaky ReLU is chosen for its advantages over other variants, addressing the "dying ReLU" issue by eliminating zero-slope segments and speeding up the training process due to its better balance, leading to faster learning[65]. It's important to note that, while leaky ReLU offers benefits, it is not necessarily superior to simple ReLU and should be considered as an alternative. The general equation for both ReLU and leaky ReLU activation functions is as follows:

$$F(x) = x^{+} = \max(0,x) \qquad \begin{Bmatrix} ZZ > 0 \\ \propto ZZ \leq 0 \end{Bmatrix} \tag{17}$$

When z is less than 0, leaky ReLU allows a small nonzero, constant gradient α.
Generally, the value of α = 0.01

Flattening :

The flattening operation, which converts image matrices into a single vector, is particularly advantageous in the context of Alzheimer's disease classification[66]. This operation streamlines the input data, enabling the creation of a fully connected first neuron layer. The subsequent fully connected layers contribute significantly to the network's capacity to discern and classify intricate patterns associated with Alzheimer's disease.

In the broader landscape of medical image classification, CNNs have demonstrated notable success in Alzheimer's disease diagnosis. Leveraging their capacity to learn hierarchical features from imaging data, CNNs provide a powerful tool for researchers and clinicians aiming to enhance the accuracy of Alzheimer's disease classification, ultimately contributing to early diagnosis and intervention strategies[67][68].

3.2.1.1 Types of CNN

3D CNN

3D Convolutional Neural Networks (CNNs) are specialized neural networks designed for processing volumetric data with spatial and temporal dimensions. The incorporation of deeper and wider architectures into 3D CNNs involves increasing the network's depth to facilitate hierarchical feature learning and expanding its width to consider a broader array of features within volumetric datasets[69]. This strategic approach aims to strike a nuanced balance, capturing intricate spatiotemporal details while accommodating the inherent complexities of 3D data[70]. Such tailored architectures find valuable applications in areas like video analysis and medical imaging, where a comprehensive understanding of the temporal evolution of three-dimensional structures is crucial for robust and accurate modeling.

VGG(VGG16,VGG19)

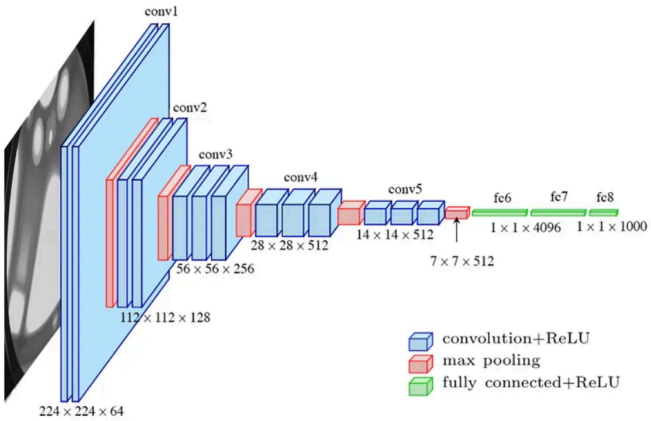


Fig 7. VGG

VGG16, crafted by the Visual Geometry Group at the University of Oxford, stands out as a widely recognized convolutional neural network architecture celebrated for its simplicity and effectiveness in image classification tasks. Boasting 16 weight layers, encompassing convolutional and fully connected layers, VGG16 follows a consistent pattern of stacking 3x3 convolutional layers succeeded by max-pooling layers[71]. This systematic approach allows it to adeptly grasp intricate features and spatial hierarchies. Its remarkable performance in the 2014 ImageNet Large Scale Visual Recognition Challenge garnered widespread acclaim. In the context of Alzheimer's disease classification, VGG16's robust capabilities have been employed for the analysis of 3D brain MRI scans, highlighting its adaptability and proficiency in identifying intricate patterns associated with distinct disease stages. Despite its computational demands, VGG16's straightforward design has established it as a benchmark in deep learning, shaping subsequent advancements in convolutional neural networks and contributing

significantly to progress in medical image analysis for Alzheimer's disease diagnosis[72].

The VGG19 model, also known as VGGNet-19, follows the same basic idea as the VGG16 model but sets itself apart by having a total of 19 layers. The numbers "16" and "19" indicate the amount of weight layers, particularly the convolutional layers, present in the model. In comparison to VGG16, VGG19 incorporates three extra convolutional layers[73]. More details about the specific features and functionalities of the VGG16 and VGG19 networks will be discussed in the following part of this conversation, providing a deeper understanding of their capabilities[74]. Additionally, it's worth noting that these networks can be applied to tasks like Alzheimer's disease classification in medical imaging.

DenseNet

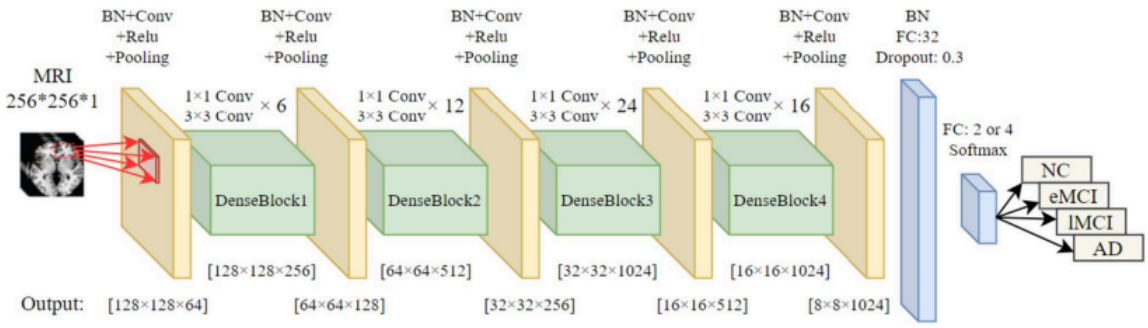


Fig 8. DenseNet 121

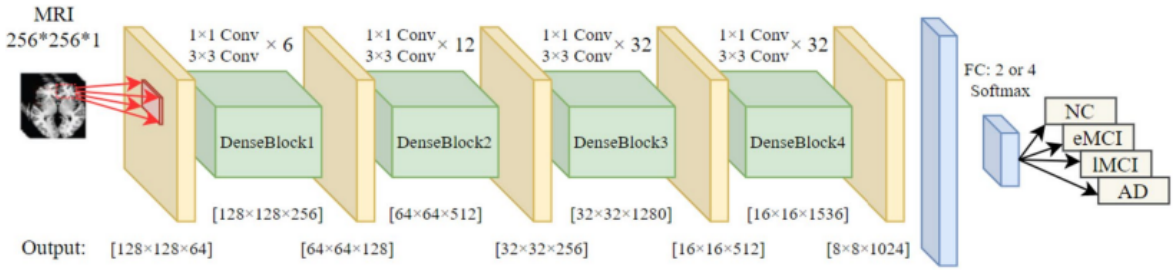


Fig 9. DenseNet 169

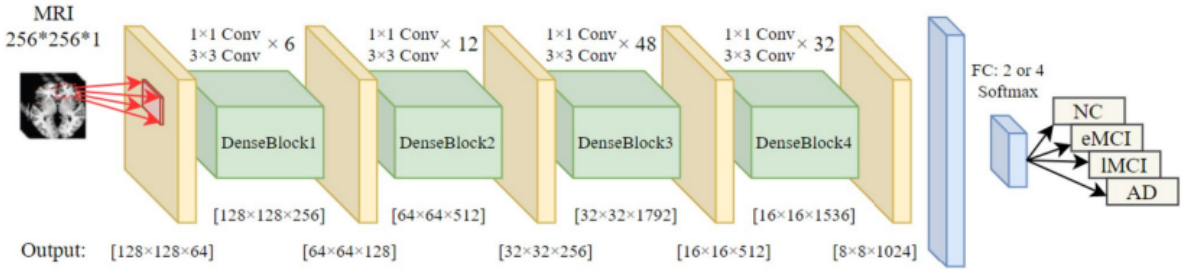


Fig 10. DenseNet 201

DenseNet architectures revolutionize neural networks by introducing a dense block structure, where each layer connects directly to every subsequent layer[75]. This design enhances information flow, addressing gradient vanishing issues. In standard DenseNet configurations, which typically consist of four dense blocks, five transition layers, one fully connected layer, and a softmax layer, intricate features are learned efficiently. Variations like DenseNet201 and DenseNet169 offer different inner layer structures. In the context of Alzheimer's disease classification, DenseNet's adaptability has been leveraged to analyze 3D brain MRI scans, demonstrating its efficacy in discerning complex patterns associated with distinct disease stages[76].DenseNet architectures, illustrated in Figure, enhance the last layer with a batch normalization layer, a fully connected layer with 32 neurons, a dropout layer with a 0.3 value, and a softmax layer, contributing to improved performance in Alzheimer's disease classification tasks.

ResNet

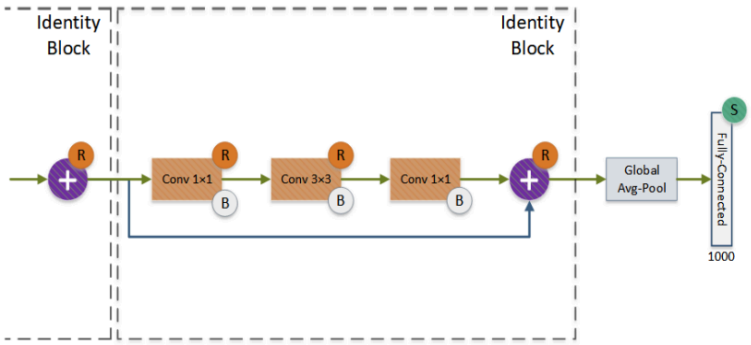


Fig 11. ResNet

ResNet, or Deep Residual Networks, revolutionizes neural network architecture by introducing shortcut connections, also known as skip connections[77]. These connections create alternative pathways for data and gradients during training, facilitating smoother flow and making training feasible. One notable ResNet model is ResNet-18, characterized by 18 layers and designed to process images with dimensions of 224×224×3. Developed by Kaiming, ResNet-18 emerged as the winner of the ILSVRC-2015 competition. Above Figure illustrates the detailed architecture of ResNet-18, showcasing how skip connections run in parallel to address the challenges of training deep neural networks[78].

Inception ResNet, combining the Inception block with the ResNet structure, offers a synergistic approach to enhance performance. The ResNet module enables the training of deeper neural networks by introducing connection residuals, while the Inception block efficiently gathers information from various image scales. This fusion not only improves CNN architecture performance but also aids in network optimization and overcoming overfitting problems[79]. With an impressive 164 layers, Inception ResNet effectively classifies around 1000 object categories. Noteworthy among ResNet variants is ResNet50, a widely used deep learning network in computer vision tasks with 50

layers. Employing convolutional layers to extract features and fully connected layers for image classification, ResNet50 proves to be an effective and efficient architecture applied across diverse applications. In the context of Alzheimer's disease classification tasks, ResNet models, including ResNet-18, ResNet50, and ResNet101, have proven instrumental in analyzing 3D brain MRI scans. The ingenious use of skip connections in ResNet architectures contributes to their adaptability and proficiency in discerning complex patterns associated with different stages of Alzheimer's disease[80]. These models, with varying depths and complexities, showcase their versatility in medical imaging applications, providing valuable insights for accurate disease diagnosis and understanding disease progression.

Inception

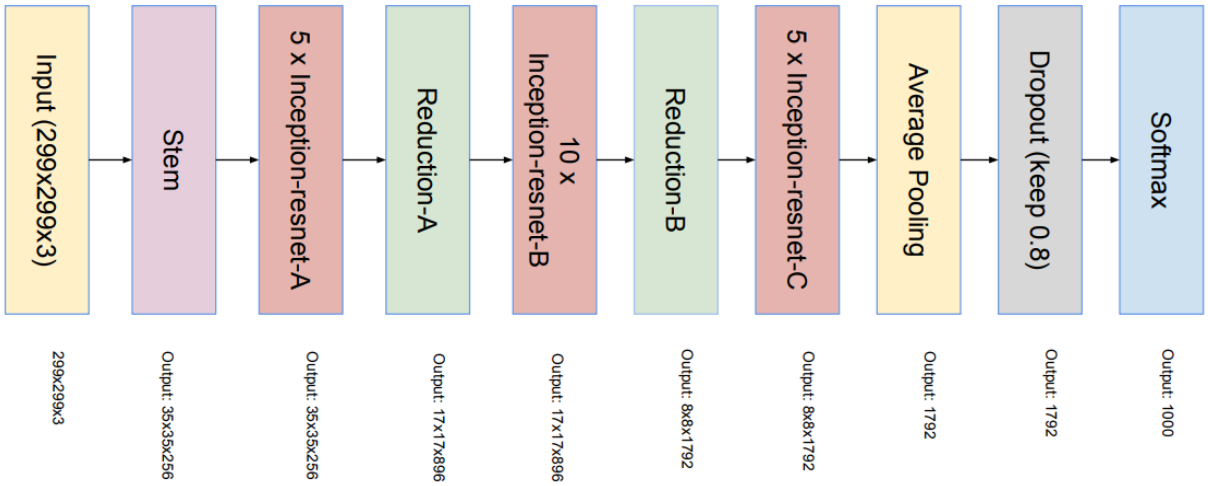


Fig 12.Inception V1, V2

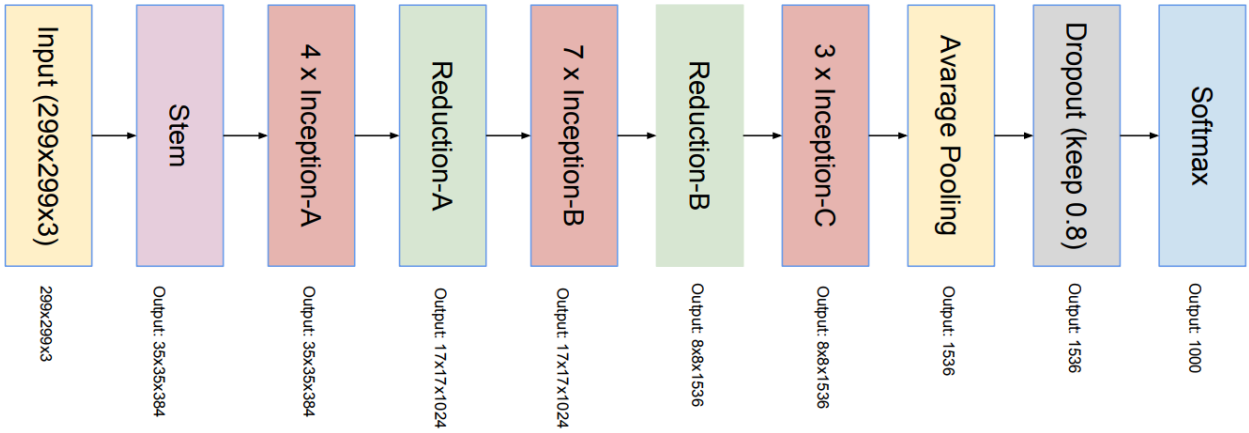


Fig 13.Inception V4

In the landscape of deep learning, the Inception architecture, also known as GoogleNet, has evolved through multiple versions, each refining its capabilities. InceptionV1, the original model, introduced the novel concept of the Inception module, which efficiently employs parallel convolutions with different kernel sizes. This pioneering design aimed to strike a balance between network width and depth, allowing for effective feature extraction at various scales[81]. Applied widely in computer vision tasks, InceptionV1 laid the foundation for subsequent iterations. Building upon the success of InceptionV1, InceptionV2, also known as Inception-ResNet, combined the Inception block with the residual connections pioneered by ResNet. This hybrid model effectively addressed challenges related to training convergence by allowing the gradients to flow directly to earlier layers[82]. The integration of Inception and ResNet structures created a powerful architecture that excelled in image classification tasks, offering a favorable trade-off between depth and width. InceptionV3 continued the legacy of its predecessors and achieved state-of-the-art results in the ImageNet Large Scale Visual Recognition Challenge[83]. This version maintained the Inception module's efficiency in resource utilization while introducing advancements for improved performance[84]. Widely adopted in computer vision applications, InceptionV3 showcased its adaptability and effectiveness in tasks such as object detection, image segmentation, and transfer learning.

Inception-v4, designed to process 2D images with three RGB channels, extended the capabilities of the Inception architecture. In the context of Alzheimer's disease classification, this adaptation involved triplicating grayscale coronal slices, augmenting the feature vector with additional details like subject age, sex, and coronal slice number[85]. With 497 layers, Inception-v4 demonstrated its versatility by replacing the original classifier for predicting Alzheimer's disease probability in brain MR images[86]. This modified architecture showcased its potential for intricate medical image analysis tasks, contributing to advancements in Alzheimer's disease research.

EfficientNet

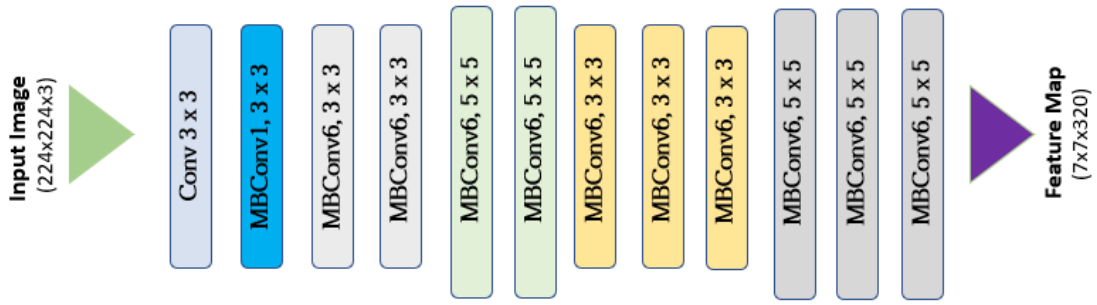


Fig 14.EfficientNet

The Convolutional Neural Network (CNN) architecture is a crucial tool for image processing, enabling the interpretation and classification of visual data. However, challenges arise in adjusting model parameters such as depth and width, leading to suboptimal configurations[87]. To address these issues, the EfficientNet Transfer Learning (TL) model utilizes a composite coefficient to evenly adjust parameters like depth, width, and resolution. This balanced optimization benchmark ensures faster training and improved accuracy.EfficientNet's core design centers around 2D Depthwise convolution units, specifically a sequence of mobile inverted bottleneck convolutions (MBConv)[88]. This innovative approach, combined with a scaling technique using predefined coefficients, distinguishes EfficientNet from traditional CNN models[89]. The unique scaling method optimizes the network's depth while preserving accuracy, making EfficientNet advantageous for various tasks, including Alzheimer's disease (AD) diagnosis.

In the realm of AD classification, the presented approach applies EfficientNet Transfer Learning techniques across different variants, including B0, B3, and B5. Each variant represents a specific balance of depth, width, and resolution adjustments, offering versatility in adapting the model to the intricacies of AD-related image data. This adaptability positions EfficientNet as a promising candidate for enhancing the accuracy and speed of AD diagnosis through medical image analysis[90][91][92].

2D CNN

In the domain of 2D Convolutional Neural Networks (CNNs), various architectures have played crucial roles, each tailored to tackle distinct challenges and objectives[93]. MobileNet, characterized by its efficiency, stands out as a CNN architecture specifically optimized for mobile devices. Its streamlined structure, featuring depthwise separable convolutions, minimizes computational requirements without compromising accuracy. This design makes MobileNet well-suited for platforms with limited resources[94].

AlexNet

AlexNet, a prominent and accurate CNN architecture, is employed as a classifier in this research, showcasing its efficacy in image processing and machine learning tasks. The architecture stands out for its strategic use of pooling layers after each convolutional layer, allowing for adaptable convolution layer sizes. Utilizing 11x11, 5x5, and 3x3 convolutions, as well as max pooling, dropout, data augmentation, ReLU activation, and dynamic SGD, AlexNet demonstrates its versatility[95]. In comparison with GoogleNet, both architectures were assessed for their performance on a test dataset of 25,000 random images, with accuracy and confusion matrices computed. AlexNet processes 224x224x3 RGB images, employing variations in convolution layers and feature maps to optimize feature extraction for effective image classification[96][97]. Subtle adjustments, such as a fully connected layer with 985 neurons, were made to enhance overall model performance.

MobileNet

MobileNet and MobileNetV2 represent Convolutional Neural Network (CNN) architectures specifically designed for optimal performance on mobile devices. MobileNet, created to address the challenges of deploying deep neural networks on resource-constrained platforms like mobile phones, is characterized by its lightweight architecture[98]. It incorporates depth wise separable convolutions, consisting of a depthwise convolution followed by a pointwise convolution, to significantly reduce computational requirements without compromising accuracy.MobileNetV2 builds upon MobileNet's foundation, introducing improvements in both architecture and performance. It features an inverse residual structure with residual links between bottleneck layers, enhancing feature learning and ensuring efficient information flow. The architecture includes an extended middle layer with lightweight depth convolutions to filter non-linear features effectively. With a full convolutional layer containing 32 filters and 19 bottleneck layers, MobileNetV2 strikes a balance between accuracy and computational efficiency[99]. Both MobileNet and MobileNetV2 exemplify advancements in CNN architectures tailored for mobile devices, offering solutions that navigate computational constraints while retaining the capacity to handle complex visual recognition tasks. In various applications, including image classification, object detection, and the context of Alzheimer's disease classification, these architectures serve as effective tools for analyzing medical images on mobile platforms.

GoogleNet

GoogleNet, also known as Inception-v1, has been trained on a vast dataset, exceeding a million images, demonstrating its ability to classify objects into 1000 categories. Originally introduced by a Google team, it achieved victory in ILSVRC-2014, presenting a 22-layer deep architecture designed for computational efficiency and practicality. Within this architecture, all convolutions, including those embedded in inception modules, leverage rectified linear unit (ReLU) activation functions. The network operates with a 224×224 receptive field size in the RGB color space, requiring image preprocessing for valid input.In the context of Alzheimer's disease, GoogleNet's versatile architecture can be employed for image classification tasks related to medical imaging, offering potential applications in the analysis of brain MRI scans for Alzheimer's diagnosis[100]. The parallel convolution lines within GoogleNet, utilizing diverse filters, could aid in capturing intricate features associated with different stages of the disease, enhancing the network's capability to discern patterns indicative of Alzheimer's progression. This adaptability underscores GoogleNet's potential contribution to advancements in medical image analysis and disease classification, including Alzheimer's.

LeNet

Introduced by Yann LeCun in 1989, LeNet stands as one of the earliest and effective Deep Neural Network (DNN) architectures, comprising a modest seven layers . In the layer arrangement post-input, it involves a convolutional layer, followed by a pooling layer, another convolutional layer, and subsequent pooling, with the sequence culminating in fully connected layers and the output layer[101]. Within Convolutional Neural Networks (CNNs), the convolutional layer plays a pivotal role, extracting essential features from input data by preserving pixel correlations.LeNet-5, a variant of the LeNet architecture. Initially recognized for its proficiency in digit classification and handwritten check number processing, LeNet-5's versatility expanded over time. Despite its historic roots, this architecture has demonstrated adaptability to complex challenges. In this context, our application involved binary classification, specifically addressing Alzheimer's and Normal data using fMRI inputs[102]. LeNet-5, chosen for its aptness, underwent adjustments to suit the intricate nature of the binary classification task associated with Alzheimer's disease.

NasNet

NasNet, short for Neural Architecture Search Network, is a deep learning architecture designed to automate the process of discovering optimal neural network architectures[103]. Employing reinforcement learning, it searches through a predefined space of neural network architectures, aiming to identify configurations that yield superior performance. In the realm of Alzheimer's disease classification, NasNet contributes by automating the design of neural networks, potentially enhancing the model's ability to discern relevant patterns and features from medical data.

TabNet

TabNet is a neural network architecture specifically tailored for tabular data, making it suitable for structured datasets commonly found in medical research[104]. It utilizes sequential attention mechanisms to focus on relevant subsets of features during training, enabling effective information extraction. In Alzheimer's disease classification, TabNet may prove beneficial for handling structured clinical data, offering interpretability and enhanced performance in discerning patterns associated with different disease stages.

SegNet

SegNet, short for Segmentation Network, is designed for semantic image segmentation tasks. It excels in delineating and categorizing specific regions within images[105]. In the context of Alzheimer's disease, SegNet may be applied to segment and analyze

medical images, such as brain MRI scans. By highlighting relevant regions associated with the disease, SegNet contributes to a more detailed understanding of structural changes, aiding in accurate classification and diagnosis.

3.2.2 RNN

With their unique ability to model sequential dependencies, RNNs prove valuable in capturing temporal patterns within diverse data types, such as time-series neuroimaging data or longitudinal patient records. The temporal dynamics inherent in Alzheimer's disease progression make RNNs well-suited for tasks involving the identification of subtle changes over time. Leveraging hidden states and memory cells, RNN architectures, including Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU), excel in learning and retaining essential information for accurate disease classification[106]. The application of RNNs in Alzheimer's research not only enhances predictive modeling but also facilitates a deeper understanding of the disease's temporal evolution, paving the way for more effective early detection and intervention strategies.

LSTM

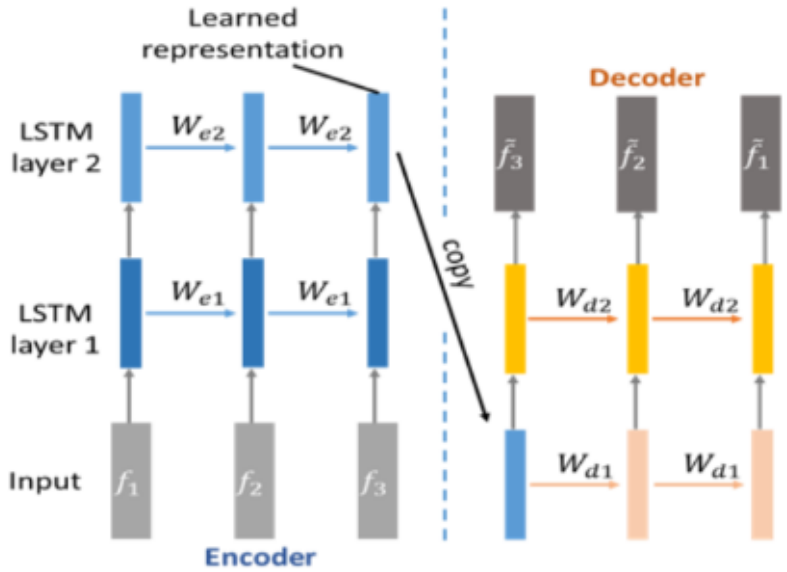


Fig 15. LSTM

Capturing longitudinal cognitive measures for each subject and understanding their overall cognitive performance and temporal changes across multiple time points is crucial. The LSTM autoencoder is a specialized tool chosen for this purpose, designed to provide informative and compact representations[107].Figure 15 illustrates the LSTM

autoencoder, consisting of an encoder and a decoder. The encoder processes input data from various time points, encoding input measures and their temporal dynamics. Conversely, the decoder reconstructs input measures sequentially based on learned representations by the encoder. The optimization process minimizes the deviation between reconstructed and input measures, resulting in a learned representation that characterizes overall cognitive performance and its dynamics. In Figure 15, LSTM autoencoder includes two LSTM layers for both the encoder and decoder. At any given time point (t=1,2,3 for illustration), f_t represents input cognitive measures, and $\tilde{f}_t$ represents the reconstructed measures. W_{ei} and W_{di} are trainable parameters for the i-th LSTM layer in the encoder and decoder, respectively, governing the forget gate, input gate, cell state, and hidden state within one LSTM layer. The optimization employs the Euclidean distance between reconstructed and input measures as the objective function.

The number of LSTM layers is chosen for generalizable performance with a minimal number of trainable parameters. The autoencoder was applied to longitudinal cognitive measures from a diverse dataset[108]. After obtaining the autoencoder, the encoder was employed on subjects with Mild Cognitive Impairment (MCI) from multiple cohorts, including ADNI-1 and GO&2. This process extracted latent features from their longitudinal cognitive measures, paving the way for subsequent prognostic analyses.

4. EVALUATION METRICS

In real-world scenarios, experts manually generate ground truth data for patients after MRI acquisition, a process prone to errors, subjectivity, and difficulty in reproducibility, even by the same expert. To address these limitations, alternative validation methods involving software simulation and phantoms have been proposed. Software simulation generates artificial images mimicking real acquisition processes, allowing for known ground truth and independent analysis of acquisition parameters and imaging artifacts. While this method is less labor-intensive and more flexible, it may not capture all factors affecting real image acquisition, providing only an approximate estimation of real images. The challenges of validation and quantitative comparison persist across different segmentation methods. The validation process necessitates ground truth data to assess the accuracy of segmented outputs. However, obtaining a sufficient amount of ground truth data for human MRI acquisitions is often a practical challenge. MRI scanners produce phantom images that are more realistic than those from software simulations, yet lack flexibility. Both software simulations and phantom imaging are resource-intensive. The evaluation of the overlap between predicted brain MRI and ground truth images involves employing various similarity measures. The most well-known evaluation measures are Recall, precision and F-Measure, Sensitivity, Specificity [109]. The performance was also measured in terms of

the classification rate and the AUC of ROC for more accurate evaluation besides recall, precision and F-Measure [110]. Let's look at these evaluation measures in detail

Recall, also called Sensitivity, is the ratio of correctly identified positive cases to the total number of actual positive cases. It's a useful metric when it's crucial to capture as many positive instances as possible, minimizing the chances of missing relevant cases. High recall indicates that the model is good at finding most of the positive examples, even if it means having more false positives.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{18}$$

Precision is a metric that evaluates the accuracy of a model's positive predictions. It is calculated as the ratio of true positives to the sum of true positives and false positives. Precision provides insights into the reliability of the model when it asserts that an instance belongs to the positive class, measuring the proportion of true positive predictions among all predicted positive instances.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{19}$$

F-measure or F1 score, is a composite metric that balances precision and recall. It is the harmonic mean of precision and recall and is particularly useful when there is an uneven distribution between the two. The F-measure aims to provide a single value that represents a trade-off between minimizing false positives and false negatives, offering a comprehensive evaluation of a model's performance.

$$\text{F-measure} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \tag{20}$$

Specificity is a metric that measures a model's ability to correctly identify negative instances. It is calculated as the ratio of true negatives to the sum of true negatives and false positives. Specificity complements recall by focusing on the accurate identification of instances belonging to the negative class, providing insights into the model's performance with respect to negatives.

$$\text{Sensitivity} = \frac{TN}{TN + FP} \tag{21}$$

AUC-ROC (Area Under the Receiver Operating Characteristic curve) quantifies the area under the ROC curve, providing a single scalar value to summarize the overall performance of a classification model. The ROC curve plots the true positive rate (sensitivity) against the false positive rate (1 - specificity) for various threshold values, demonstrating the model's ability to distinguish between the positive and negative classes.

Table shows the different papers we collected along with the information like the dataset used, proposed methodology and the accuracy that method gained. We included the dataset information which is based on classes (AD, CN, MCI, pMCI, sMCI).

Papers	AD	CN	MCI	pMCI	sMCI	Dataset Used	Method	Accuracy
1. Voxel-wise Fusion of Resting fMRI Networks and Gray Matter Volume for Alzheimer's Disease Classification using Deep Multimodal Learning [111].	83	383	264	-	-	ADNI	3DCNN	AD vs CN: 94.12% AD vs CN vs MCI: 62.68%
2. A Novel Framework for Classification of Different Alzheimer's Disease Stages Using CNN Model [62].	145	-	198	-	-	ADNI	MobileNet model	CN vs. MCI vs. EMCI vs. LMCI vs. AD: 96.22%
3. Deep Learning-Based Ensembling Technique to Classify Alzheimer's Disease Stages Using Functional MRI [86].	1.35 k	1.35k	1.35k	-	-	ADNI	Ensembling of VGG-16, ResNet-18, AlexNet, Inception V1, and CustomCN N	AD vs CN: 99.6 MCI vs AD: 99.4 CN vs MCI: 99.8
4. Deep Multi - Branch CNN Architecture for Early Alzheimer's Detection from Brain MRIs [59].	180	-	115	-	-	ADNI	Deep CNN	MCI vs AD: 83%

5. The Application of Deep Learning for Classification of Alzheimer's Disease Stages by Magnetic Resonance Imaging Data [60].	453	748	981	-	-	ADNI	3D CNN	MCI vs AD: 93%
6. A modified 3D EfficientNet for the classification of Alzheimer's disease using structural magnetic resonance images. [8]	182	218	-	80	99	ADNI	3D EfficientNet	CN vs AD: 95.00%, CN vs pMCI: 86.67%, sMCI vs pMCI: 83.33%
7. Generalizable deep learning model for early Alzheimer's disease detection from structural MRIs [112].	748	782	1089	-	-	ADNI, NACC	3D CNN, ROI-volume /thickness model	AD vs CN: 93% - ADNI, MCI vs CN: 88% - ADNI, AD/MCI vs CN: 95% - ADNI
8. Convolutional Neural Networks for Classification of Alzheimer's Disease: Overview and Reproducible Evaluation. [38]	336	330	787	295	298	ADNI, AIBL,OASIS	3D subject-level, ROI-based,patch-level CNN, 2D slice-level CNN and Linear SVM	SVM: AD vs CN: 88% ± 0.02 sMCI vs pMCI: 70% ± 0.06 3D patch-level CNN:AD vs CN: 81% ± 0.03 sMCI vs pMCI: 75% ± 0.04
9. Explainable AI-based Alzheimer's prediction and management using multimodal data. [30]	669	2248	-	-	-	OASIS	SVM	98.81%

10. Reproducible evaluation of diffusion MRI features for automatic classification of patients with Alzheimer's disease. [37]	46	46	97	24	54	ADNI	Linear SVM	CN vs AD: 76% CN vs MCI: 60% CN vs pMCI: 61% sMCI vs pMCI: 51%
11. Multi-view Classification for Identification of Alzheimer's Disease. [9]	198	229	403	124	118	ADNI	SVM	AD vs NC: 91.31% MCI vs NC: 78.07% pMCI vs sMCI: 75.54%
12. Alzheimer's disease detection using convolutional neural networks and transfer learning based methods [113].	86	80	-	-	-	OASIS	CNN, Transfer Learning	92.86% (Transfer Learning)
13. Dementia classification from magnetic resonance images by machine learning [58].	200	-	400	-	-	OASIS	SVM, KNN, Naive Bayes, CNN, VGG-16 ResNet-50	SVM: 89.9% ± 3.2
14. A multi-stream convolutional neural network for classification of progressive MCI in Alzheimer's Disease using structural MRI images [8].	200	231	-	164	100	ADNI	multi-stream deep CNN	AD vs CN: 97.78% pMCI vs sMCI: 79.90%
15. Automated Detection of Alzheimer's Disease and Mild Cognitive Impairment Using Whole Brain MRI [67].	163	163	163	-	-	ADNI	CNN	AD vs CN: 97% MCI vs CN: 96% AD vs MCI: 90%

16. Automatic Classification of Alzheimer's Disease using brain MRI data and deep Convolutional Neural Networks [43].	200	200	-	-	-	ADNI	3D ResNet-20	AD vs CN: 93.75%
17. Pre-Training 3D Convolutional Neural Networks For Prodromal Alzheimer's Disease Classification [68].	332	486	173	264	331	ADNI	CNN	AD vs CN: 70.8% ± 3.6
18. A Binary Classifier Using Fully Connected Neural Network for Alzheimer's Disease Classification [114].	58	60	60	-	-	ADNI	CNN	AD vs MCI: 83.33% AD vs CN: 87.50% MCI vs CN: 79.17%
19. Alzheimer and Mild Cognitive disease Recognition Using Automated Deep Learning Techniques [104].	39	45	68	-	-	ADNI	CNN	AD vs CN vs MCI: 99.3%
20. Classification of Alzheimer's disease based on several features extracted from MRI T1-weighted brain images [115].	200	228	407	-	-	ADNI	SVM	AD vs CN: 93.3% using SVM
21. Alzheimer's Disease Early Detection Using Machine Learning Techniques [116].	471	529	-	-	-	ADNI, OASIS	SVM	AD vs CN: 99.10%
22. Classification of Initial Stages of Alzheimer's Disease through Pet Neuroimaging Modality and Deep Learning: Quantifying the Impact of Image Filtering Approaches [89].	94	102	97	-	-	ADNI	3D CNN	AD vs CN: 86.22% (using Box Filtering)

23. Classification of multimodal MRI images using Deep Learning : Application to the diagnosis of Alzheimer's Disease [117].	64	399	273	-	-	ADNI	CNN	AD vs CN: 92.30% AD vs MCI: 79.16% MCI vs CN: 78.48%
24. Deep Learning Approach for Early Detection of Alzheimer's Disease [93].	75	75	-	-	-	ADNI	2D CNN, 3D CNN	Proposed 2D-M2 IC model: AD vs. NC: 97.11% Proposed 3D-M2 IC model: AD vs CN: 97.36%
25. Classification of Alzheimer's Disease and Mild Cognitive Impairment Based on Cortical and Subcortical Features from MRI T1 Brain Images Utilizing Four Different Types of Datasets [118].	32	28	-	-	-	ADNI, NRCD, ARWIBO , NACC	RBF-SVM	AD vs CN: ADNI: 91.57% NRCD: 97.37% NACC: 95.24% ARWIBO: 94.74%
26. Deep learning-based classification of multi-categorical Alzheimer's disease data [119].	153	268	321	-	-	ADNI	1D CNN, ANN	Overall Accuracy: 1D CNN: 88.275% 87.197% for ANN
27. Using Machine Learning Methods for Detecting Alzheimer's Disease through Hippocampal Volume Analysis [120].	159	109	217	-	-	ADNI	KNN, GNB, SVM, LR, DT, RF	KNN - 80% GNB - 82% SVM - 80%

28. 3D CNN-based classification using sMRI and MD-DTI images for Alzheimer disease studies [121].	48	48	108	-	-	ADNI	3D CNN	AD vs CN: 96.7% AD vs MCI: 80% MCI vs CN: 65.8%
29. An Efficient 3D Deep Convolutional Network forAlzheimer's DiseaseDiagnosis using MR Images [122].	199	141	-	-	-	ADNI	3D ConvNet architecture	AD vs CN: 98.74%
30. Deep Learning on MRI Affirms the Prominence of the Hippocampal Formation in Alzheimer's Disease Classification [123].	124	120	138	311	-	ADNI	3D CNN	AD vs CN: 93.3%
31. Classification of Alzheimer's Disease Structural MRI Data by Deep Learning Convolutional Neural Networks [101].	211	91	-	-	-	ADNI	CNN	AD vs CN: 96.05%
32. Alzheimer's Disease Diagnostics by a Deeply Supervised Adaptable 3D Convolutional Network [99].	70	70	70	-	-	ADNI	DSA 3D CNN	AD vs CN: 99.3% AD vs MCI: 100% MCI vs CN: 94.22%
33. Discriminative multi-task feature selection for multi-modality classification of Alzheimer's disease [124].	51	52	99	-	-	ADNI	SVM	AD vs CN: 95.92% MCI vs CN: 82.13%
34. Classification of Alzheimer's Disease from MRI Using Sulcal Morphology [125].	110	100	-	-	-	ADNI	SVM	AD vs CN: 87.50%

35. Independent Component Analysis-Based Classification of Alzheimer's Disease from Segmented MRI Data [126].	188	229	401	-	-	ADNI	SVM	AD vs CN: 90.4%

5. RESULTS AND DISCUSSION

In the realm of Alzheimer's disease classification, the exploration of various machine learning (ML) and deep learning (DL) models in recent research papers unveils an interesting landscape. Traditional ML models, particularly Support Vector Machines (SVM), stand out as notable competitors, showcasing admirable accuracy levels. SVMs, with an accuracy of 87.5%, reveal their robust capability in recognizing patterns within medical imaging data, especially excelling in tasks demanding well-defined decision boundaries[127].On a parallel track, the forefront of deep learning models is dominated by Convolutional Neural Networks (CNN), representing an advanced pattern in Alzheimer's disease classification. CNNs exhibit exceptional effectiveness, having an extraordinary accuracy of 99.8%[128]. Their standout feature lies in the automatic extraction of features from raw data, enabling them to capture complex spatial hierarchies and refined details embedded in medical images. Despite the challenges faced in capturing complex patterns within high-dimensional data, SVMs maintain their status as solid performers. Demonstrating reliability in scenarios with well-defined decision boundaries, SVM achieves accuracies of 83.27% and 79% in specific studies, addressing tasks related to retinal vasculature and 2D brain images, respectively[129][130]. Shifting the focus to the deep learning paradigm, CNNs emerge as the best one in studies where accuracies of 96% and 92.19% were achieved[131][132]. The remarkable performance of CNNs is attributed to their inherent ability to automatically uncover hierarchical features, providing a crucial advantage in understanding complex patterns within fMRI data.

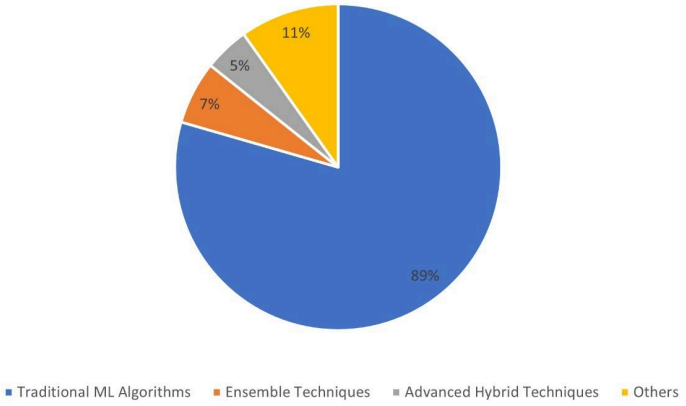


Fig 16. Different Machine Learning Models

In a broader examination of the generality of Machine Learning (ML) and Deep Learning (DL) models in recent Alzheimer's disease research, the accompanying pie charts offer illuminating insights into the landscape of these methodologies. Traditional ML models, with Support Vector Machines (SVM) taking the lead, comprise a substantial 89% of the distribution. This prominence underscores the enduring popularity of well-established ML algorithms in the context of Alzheimer's disease classification.

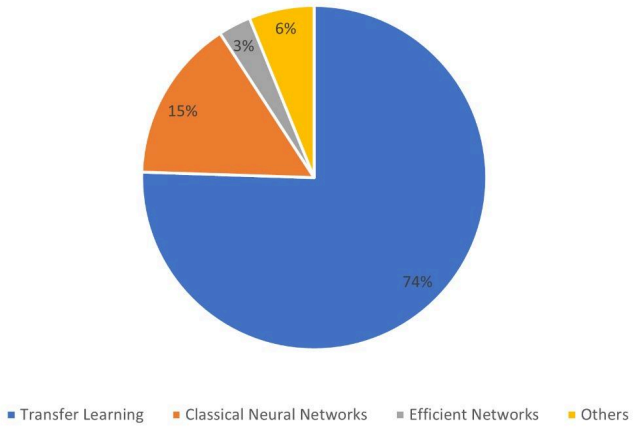


Fig 17. Different Deep Learning Models

Within the Deep Learning landscape, Convolutional Neural Networks (CNNs) serve as central figures. This signifies their substantial role in advancing diagnostic methodologies, particularly in the realm of medical imaging and Alzheimer's disease research. CNNs, renowned for their capability to automatically extract features from raw data, showcase their pivotal contribution to the evolving landscape of deep learning applications in healthcare. Transfer learning, encompassing a significant 74% of the distribution, emerges as a dominant technique in DL. This occurrence indicates a clear trend in utilizing pre-trained models, including CNN architectures such as ResNet, VGG, Inception, and others, to enhance the efficiency and effectiveness of DL algorithms for tasks related to Alzheimer's disease. Transfer learning's widespread adoption underscores the practicality of building upon existing knowledge and expertise in the field, particularly in the context of complex medical datasets. In conclusion, the comprehensive analysis of the distribution of ML and DL models, as evidenced by the pie charts, provides valuable insights into the evolving landscape of Alzheimer's disease classification. This collective understanding paves the way for the development of more effective and sophisticated diagnostic approaches, bridging the gap between traditional and advanced methodologies in the pursuit of enhanced healthcare solutions for Alzheimer's disease.

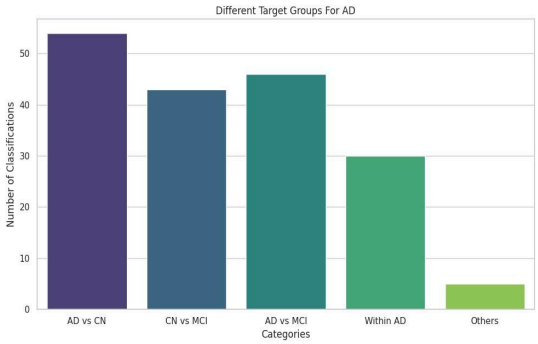


Fig 18. Machine Learning Bar Graph

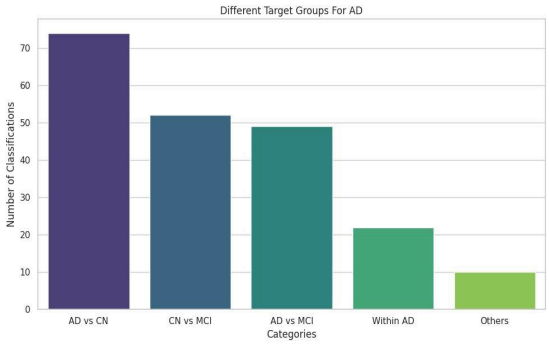


Fig 19.Deep Learning Bar Graph

In the Alzheimer's dataset(ADNI, OASIS), there are different categories or targets. One important target is distinguishing between Mild Cognitive Impairment (MCI) and Alzheimer's Disease (AD), which is crucial for early AD diagnosis. By looking at graphs Fig 18 and Fig 19, we notice that most of the research focuses on telling apart Cognitively Normal (CN) from AD and CN from MCI. But there's not much research on MCI vs AD, which is important for early AD detection. There are also fewer studies on categories like MCIc vs MCInc and MCIc vs MCIs. So, in the future, researchers can pay more attention to these specific issues for better early detection of dementia caused by Alzheimer’s disease.

Another important task is to ensure that we have the right data. It's crucial to find accurate and relevant information for our research. In the context of Alzheimer's disease diagnosis, a key focus is on developing personalized diagnosis models tailored to each individual. To achieve this, we can utilize diverse clinical data sources that match the characteristics of different populations. Moreover, it's essential to create innovative learning techniques, especially when dealing with small datasets in real-world scenarios where sample sizes may be limited for model training. Additionally, as Alzheimer's data collection often contains noise from various sources, employing techniques that are insensitive to this noise is essential for accurate classification.

6. CONCLUSIONS

The field of medical imaging has witnessed significant advancements with the integration of computational intelligence, deep learning, and computer-aided detection, particularly in the context of brain segmentation and Alzheimer's disease (AD) diagnosis. While deep learning methods have shown substantial impact on the quantitative analysis of brain MRI, developing a robust and generic approach remains challenging. The effectiveness of deep learning techniques is influenced by pre-processing initialization and post-processing, highlighting the need for careful consideration in these stages. This review explores studies focusing on brain structure and the classification of brain MRI for AD diagnosis. A key aspect discussed is how brain structure segmentation contributes to enhancing the classification performance of AD. The segmentation of brain MRI plays a crucial role in facilitating the interpretation and classification of AD, addressing challenges such as noisy background, partial volume effect, and low contrast inherent in the images. The automatic classification of AD faces additional hurdles, primarily stemming from the low contrast of anatomical structures in MRI scans. In response to these challenges, various segmentation methods with varying complexities have been proposed over the past few decades, leading to improved accuracy. Notably, the application of deep learning approaches in brain structure segmentation and AD classification has gained attention due to their ability to deliver effective results across large-scale datasets and their capacity to independently learn and make decisions. This shift towards leveraging deep learning signifies a promising direction for advancing the accuracy and reliability of AD diagnosis through medical imaging. In this work, papers using machine learning techniques and deep learning techniques are analyzed for diagnosis of Alzheimers. Research on other learning techniques like transfer, ensemble, and multi-kernel learning is also discussed. This can be useful for researchers working on any of these techniques to work on Alzheimers. As per this survey, it can be stated that SVM based models have been widely used for Alzheimer’s disease showing its robustness. This is because techniques like CNN suffer from the drawbacks of local minima, which is not the case with SVM. However, CNNs are more versatile and robust when it comes to incremental learning, modeling sequential data, and quantizing high dimensional spaces. Therefore, novel variants of CNN can be used for Alzheimer‘s in future. Deep learning and ensemble learning techniques give promising results by modeling highly complex data with high accuracy. The abundant usage of SVM also stems from the fact that it is easier to interpret as compared to deep neural networks which act as black box models. This problem should be addressed in future by focusing on clinical interpretability of deep learning models. Also, more work is needed in proper integration of feature selection techniques and machine learning models for a particular modality of data. It is also observed that researchers have given more importance to the feature extraction phase, and not much to the classification phase. This can be addressed in future research, since novel models can give some new insight in the diagnosis of Azheimers. Moreover, more work is required in formulation of machine learning models which can integrate information from various modalities for early diagnosis of Alzheimer’s disease.

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